Explainable Shipment Route Optimizer

Method, architecture, and measured outcomes · Assignment prototype

Problem and scope

RouteWise compares candidate transport routes where spend, delivery time, and risk trade off. It is a local decision-support demonstration, not a carrier forecast or production recommendation. No machine-learning model is trained.

The fixed-seed synthetic dataset contains **30 shipments** and **120 candidate routes** (four per shipment). **20** shipments have at least one route meeting all limits; **10** have none. There are **36** computed-feasible route records.

System design

A React and TypeScript dashboard calls typed FastAPI endpoints. Pydantic validates API inputs; the Python route optimizer is the single authority for computed feasibility, normalization, scoring, ranking, and explanations. A separate evaluation service reuses that engine for profile sensitivity and feasible-route baselines. The CSV files are loaded using paths relative to the project.

Core safety rule: stored CSV feasibility flags are audit targets, not the selection source. All five mandatory limits are recomputed, and a shipment with no eligible route returns no recommendation plus candidate-specific rejection reasons.

Optimization methodology

For each candidate, cost (USD), transit days, delay probability, congestion risk, and geopolitical risk must all be no greater than the shipment's corresponding maximum. Invalid or missing values are rejected. Optional request overrides replace supported maxima for that call.

Objective

Cost and time are min-max normalized among computed-feasible routes for the same shipment. Equal values map to neutral 0.5. Risk passes raw unit interval weights through. The lower penalty ranks first.

Risk = 0.50 × delay + 0.25 × congestion + 0.25 × geopolitical risk

Penalty = w_{cost} × normalized cost + w_{time} × normalized time + w_{risk} × risk

The coefficients are illustrative assumptions, not estimates learned from observed outcomes. Weights are finite, nonnegative, and sum to one within 1e-9. Presets are cost-sensitive, balanced (0.40/0.30/0.30), time-sensitive, and reliability-sensitive. Route ID breaks equal displayed-score ties deterministically.

Evaluation protocol

The weighted optimizer is compared with the cheapest, fastest, and lowest-risk computed-feasible candidates. Each selected route is evaluated with the balanced objective for comparison. Feasibility rate uses all 30 shipments as its denominator. Pareto analysis minimizes cost, duration, and illustrative risk independently for each shipment.

A baseline is a selection rule, not ground truth. A lower score or a lower individual metric on these generated records does not establish real-world optimality, predictive accuracy, or savings.

Measured results and limitations

Balanced-profile evaluation: 30 shipments, 20 with an eligible route, 10 with none. Feasibility label mismatches: 0. Every reported strategy has zero constraint violations.

Strategy	Avg cost (USD)	Avg time (days)	Avg risk	Avg balanced penalty
Weighted	17,111.63	19.25	0.397475	0.414555
Cheapest	14,874.19	20.34	0.425638	0.420191
Fastest	23,946.93	17.09	0.396181	0.449167
Reliability	19,831.00	18.71	0.379493	0.437946

Against balanced, profile changes were 2 cost-sensitive, 7 time-sensitive, and 3 reliability-sensitive selections out of 20 feasible shipments. The per-shipment Pareto frontiers contain 34 of 36 feasible routes (average size 1.7).

Hypothetical disruption analysis

The Disruption Lab applies bounded deterministic risk increments and cost/time percentage changes to copied route records, then reuses the optimizer to recompute feasibility and rankings. It reports per-route metric changes, rejections, feasible counts, recommendation switches, and fallback availability. The source dataset is not mutated; if no candidate passes the active limits, the result has no recommendation. This is illustrative scenario analysis, not a forecast or measured resilience claim.

Interpretation and limits

In this snapshot, the weighted strategy's mean balanced penalty is 0.414555; the cheapest baseline has lower mean spend but a higher mean balanced penalty and longer mean transit time. The fastest baseline averages 17.09 days at higher mean cost. The reliability-oriented baseline has the lowest mean illustrative risk penalty. These are descriptive trade-offs, not business outcomes.

The dataset is small, synthetic, and fixed-seed; profiles and risk coefficients have not been calibrated. Capacity, schedules, emissions, uncertainty, and live disruptions are outside the model. Validate constraints and scoring with logistics stakeholders and governed operational data before any real use. Future evaluation should include observed disruptions, freshness controls, human review, and additional stakeholder-approved objectives.